

Obliczenia Naukowe

Ćwiczenia

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1 Lista 1

1.1

$$\begin{aligned}
 a_1 &= 0,25 \quad \tilde{a}_1 = 0,25 * 10^0 \\
 a_2 &= 0,0046 \quad \tilde{a}_2 = 0,46 * 10^{-2} \\
 a_3 &= 0,00079 \quad \tilde{a}_3 = 0,79 * 10^{-3} \\
 a_4 &= 0,061 \quad \tilde{a}_4 = 0,61 * 10^{-1} \\
 ((\tilde{a}_1 \oplus \tilde{a}_2) \oplus \tilde{a}_3) \oplus \tilde{a}_4 \\
 \tilde{a}_1 \oplus \tilde{a}_2 &= 0,25 * 10^0 \oplus 0,46 * 10^{-2} = 0,25 * 10^0 \oplus 0,0046 * 10^0 = rd(0,2546 * 10^0) = \\
 &0,25 * 10^0 = \tilde{a}_{12} \\
 \tilde{a}_{12} \oplus \tilde{a}_3 &= 0,25 * 10^0 \oplus 0,79 * 10^{-3} = rd(0,25079 * 10^0) = 0,25 * 10^0 = \tilde{a}_{123} \\
 \tilde{a}_{123} \oplus \tilde{a}_4 &= 0,25 * 10^0 \oplus 0,61 * 10^{-1} = rd(0,311 * 10^0) = 0,31 * 10^0 \\
 ((\tilde{a}_3 \oplus \tilde{a}_2) \oplus \tilde{a}_4) \oplus \tilde{a}_1 \\
 \tilde{a}_3 \oplus \tilde{a}_2 &= 0,54 * 10^{-2} \\
 \tilde{a}_{32} \oplus \tilde{a}_4 &= 0,66 * 10^{-1} \\
 \tilde{a}_{432} \oplus \tilde{a}_1 &= 0,32 * 10^0 \\
 \text{Wynik rzeczywisty} &= 0,31639 \\
 \delta &= \frac{|0,31639 - 0,31|}{|0,31369|} = 0,0201966 \\
 \delta &= \frac{|0,31639 - 0,32|}{|0,31369|} = 0,01141
 \end{aligned}$$

1.2

1.2.a)

$$\begin{aligned}
 x &= 0,54617 \quad \tilde{x} = 0,5462 \\
 y &= 0,54601 \quad \tilde{y} = 0,5460 \\
 r &= x - y = 0,00016 \quad \tilde{r} = \tilde{x} \ominus \tilde{y} = rd(0,00016) = 0,2 * 20^{-3} \\
 \epsilon &= 0,5\beta^{1-t} = 0,5 * 10^{-3}
 \end{aligned}$$

1.2.b)

$$\begin{aligned}
 \beta &= 10 \quad t = 3 \\
 a &= 1,22
 \end{aligned}$$

$$\begin{aligned}
b &= 3, 34 \\
c &= 2, 28 \\
\tilde{\Delta} &= \tilde{b} \odot \tilde{b} \ominus 4 \odot \tilde{a} \odot \tilde{c} = rd(3, 34 * 3, 34) \ominus \dots = 11, 16 \ominus 4 \odot \tilde{a} \odot \tilde{c} = 0, 4 * 10^1 \odot \\
&0, 122 * 10^1 = (0, 488 * 10^{-1}) \\
&(0, 488 * 10^1) \odot (0, 228 * 10^1) = (0, 111 * 10^2) \\
\Delta &= (0, 112 * 10^1] \ominus (0, 111 * 10^2) = (0, 001 * 10^2) = 0, 1 \\
\Delta &= 0, 0292 \\
\frac{|\Delta - \tilde{\Delta}|}{|\Delta|} &= \frac{|0, 0292 - 0, 1|}{|\Delta|} = 2, 42 \\
\epsilon &= \frac{1}{2} \beta^{1-t} = \frac{1}{2} * 10^{-2} = 0, 005
\end{aligned}$$

1.3

$$\begin{aligned}
(0, 78125)_{10} &= (0, 11001)_2 \\
z &= \pm m_z 2^c m_z \in [\frac{1}{2}, 1) \\
0, 11001 * 2^0 \\
0, 101111001 * 2^{10} \\
(754)_{10} &= (1011110010)_2
\end{aligned}$$

1.4

1.4.a)

$$\begin{aligned}
t &= 23 \\
n &\in [1, 2) \\
x &= 2^{-1} + 2^{-26} \\
2^{-1} &= (\frac{1}{2})_{10} = (0, 1)_2 \\
2^{-26} &= (\frac{1}{2^{26}})_{10} = (0, \underbrace{0 \dots 0}_{25} 1)_2 \\
x &= 0, 1 \underbrace{0 \dots 0}_{24} 1 \\
\tilde{x} &= 1, \underbrace{0 \dots 0}_{24} 1 * 2^{-1} \text{ Odp. Nie}
\end{aligned}$$

1.4.b)

$$\begin{aligned}
y &= \frac{1}{3} = 0, (3) \\
\tilde{y} &= 1, (01)_2 * 2^{-2} \\
0, (3) &= 0, (01)_2 \\
0, (3) * 2 \\
0, (6) * 2 \\
1, 3 * 1 \text{ Odp. Nie}
\end{aligned}$$

1.4.c)

$$\begin{aligned}
z &= \frac{1}{5} = 0, 2 \\
\tilde{z} &= 1, (0011) * 2^{-3} \text{ Odp. Nie}
\end{aligned}$$

1.4.d)

$$j = \frac{1}{10} = (0,1)_{10}$$

$$0,1 * 2$$

$$\tilde{j} = 1, (00011)_2 * 2^{-4}$$

1.5

33-bitowe słowo $x = sm2^c$
cecha 8 bitów ze znakiem
mantysa 24 bity $\in [\frac{1}{2}, 1)$

1.5.a)

$$x_{max} = (0.11...1)_2 * 2^{127} = (1 - 2^{-24}) * 2^{127} = 1,7 * 10^{38}$$

$$x_{min} = (0.10...0)_2 * 2^{-127} = 2^{-1} * 2^{-127} = 2^{-128}$$

$$[-x_{max}, -x_{min}] \cup [x_{min}, x_{max}]$$

1.5.b)

$$(-x_{min}, x_{min})$$

1.5.c)

$$\epsilon = \frac{1}{2} * \beta^{1-t} = 2^{-1} * 2^{-23} = 2^{-24}$$

Single**1.5.a)**

$$x_{max} = (2 * 2^{-23}) * 2^{127} \approx 3,4 * 10^{38}$$

$$x_{min} = 1 * 2^{-126} = 2^{-126}$$

$$x_{minsub} = (0.00....01)_2 * 2^{-126} = 2^{-23} * 2^{-126} = 2^{-149}$$

$$[-x_{max}, -x_{min}] \cup \{0\} \cup [x_{min}, x_{max}]$$

1.5.b)

$$(-x_{minsub}, x_{minsub})$$

1.5.c)

$$\epsilon = \frac{1}{2} \beta^{1-t} = 2^{-1} * 2^{-23} = 2^{-24}$$

1.6

1.6.a)

$$x = 1 + 2^{-24}$$

$$\epsilon = 2^{-23}$$

$$x^- = 1$$

$$x^+ = 1 + 2^{-23}$$

1.6.b)

$$x \oplus 1 = 1$$

$$x \in (0, 2^{-24} + 2^{-25}]$$

1.6.c)

$$x \oplus = x$$

$$x < \frac{x}{2^{23}}$$

1.7

$$A(a_1, \dots, a_n) = (\dots(a_1 \oplus a_2) \oplus a_3) \oplus \dots \oplus a_n = (\dots(a_1 + a_2)(1 + \delta_1) + a_3)(1 + \delta_2) + \dots + a_n)(1 + \delta_{n-1}) = a_1 + a_1 * E_1 + a_2 + a_2 * E_1 + a_3 + a_3 * E_2 \dots + a_n + a_n * E_{n-1} = (a_1 + a_2 + \dots + a_n) + (a_1 * E_1 + a_2 * E_1 + \dots + a_n * E_{n-1}) = S + (a_1 * E_1 + a_2 * E_1 + \dots + a_n * E_{n-1})$$

$$|\delta| \leq \epsilon$$

$$1 + E_k = \prod_{i=1}^k (1 + \delta_k) \leftarrow \text{wprowadzamy}$$

$$(a_1 * E_1 + a_2 * E_1 + \dots + a_n * E_{n-1}) = E_{max} = \prod_{i=1}^{n-1} (1 + |\delta_i|) - 1 \leq \prod_{i=1}^{n-1} (1 + \epsilon) - 1$$

$$a_1 * E_1 + a_2 * E_1 + \dots + a_n * E_{n-1} \leq (a_1 + a_2 + \dots + a_n) * E_{max} \leq ((a_1 + a_2 + \dots + a_n) * \prod_{i=1}^{n-1} (1 + \epsilon)) - 1 = S(1 + \epsilon)^{n-1} - 1$$

$$\frac{|\tilde{S} - S|}{|S|} \leq \frac{\cancel{S} + S * (1 + s)}{|\tilde{S}|} \leftarrow \text{Zmazał za szybko}$$

$$\tilde{S} \leq S + S * (1 + \epsilon)^{n-1} - 1$$

1.8

$$P(n-1) = \prod_{i=0}^{n-1} q_i \text{ — dokładny wynik}$$

$$Q(n-1) = \text{fl}(P(n-1)) \text{ — wynik w fl}$$

$$Q(n-1) = P(n-1)(1 + \sum_{i=1}^{n-1} q_i), \forall i |\delta_i| \leq \epsilon \quad n=1 \quad Q(1) = a_0 \odot a_1 = a_0 a_1 (1 + \delta_1) =$$

$$P(1)(1 + \sum_{i=1}^1 q_i)$$

$$\text{Zachodzi dla } 1$$

$$Q(n+1) = a_0 \odot a_1 \odot \dots \odot a_{n+1} = Q(n) \odot a_{n+1} = (Q(n) a_{n+1})(1 + \delta_n + 1) = P(n)(1 + \sum_{i=1}^n a_{n+1}(1 + \delta_{n+1})) = P(n+1)(1 + \sum_{i=1}^n \delta_1)(1 + \delta_{n+1}) \approx P(n+1)(1 + \delta_{n+1} + \sum_{i=1}^n \delta_i) = P(n+1)(1 + \sum_{i=1}^{n+1} \delta_i)$$

$$|P(n)-Q(n)|=|P(n)\sum_{i=1}^n\delta_i|$$

1.9

$$A1(a,b)=a^2-b^2$$

$$fl(a^2-b^2)=fl(fl(a^2)-fl(b^2))=(a^2(1+\delta_1)-b^2(1+\delta_2))(1+\delta_3)\approx a^2-b^2+(a^2-b^2)\delta_3+a^2\delta_1-b^2\delta_2$$

Błąd względny:

$$\frac{|a^2-b^2+(a^2-b^2)\delta_3+a^2\delta_1-b^2\delta_2-(a^2-b^2)|}{|a^2-b^2|}=\frac{|(a^2-b^2)\delta_3+a^2\delta_1-b^2\delta_2|}{|a^2-b^2|}\leq|\delta_3|+\frac{|a^2\delta_1-b^2\delta_2|}{|a^2-b^2|}\leq|\delta_3|+\frac{a^2|\delta_1|-b^2|\delta_2|}{|a^2-b^2|}\leq\epsilon+\frac{a^2\epsilon+b^2\epsilon}{|a^2-b^2|}=\epsilon(1+\frac{a^2+b^2}{|a^2-b^2|})$$

$$A2(a,b)=(a-b)(a+b)$$

$$((a-b)(1+\delta_1)(a+b)(1+\delta_2))(1+\delta_3)=(a^2-b^2)(1+\delta_1)(1+\delta_2)(1+\delta_3)\approx(a^2-b^2)(1+\delta_1+\delta_2+\delta_3)$$

Błąd względny:

$$\frac{|(a^2-b^2)(1+\delta_1+\delta_2+\delta_3)-(a^2-b^2)|}{|a^2-b^2|}=|\delta_1+\delta_2+\delta_3|\leq|\delta_1|+|\delta_2|+|\delta_3|\leq 3\epsilon$$

1.10

$$A_1(a_0,\dots,a[n],x)-$$

2 Lista 2

2.1

$$y=\sqrt{x^2+1}-1$$

TW z wykładu:

Jeśli x,y - dodatnie liczby w dwójkowej arytmetyce float, takie że:

$$x>y, 2^{-q}\leq 1-\frac{y}{x}$$

to przy odejmowaniu tracimy najwyżej q bitów q = 2

$$2^{-2}\leq 1-\frac{1}{\sqrt{x^2+1}}$$

$$\frac{1}{4}=1-\frac{1}{\sqrt{x^2+1}}$$

$$\frac{1}{\sqrt{x^2+1}}\leq\frac{3}{4}$$

$$\sqrt{x^2+1}\geq\frac{4}{3}$$

2.2

Wskazówka:

Twierdzenie z 1 zadania: $2^{-q}\leq 1-\frac{a}{b}\leq 2^{-P}\wedge b-a, b>a$

$$x=\frac{1}{2}$$

$$\cos\frac{1}{2}=0,8775825$$

$$2^{-q}\leq 1-\frac{0,8775825}{1}\leq 2^{-P}$$

$$2^{-q}\leq 0,1224175\leq 2^{-P}$$

$$q=4\wedge P=3$$

2.3

$$f(x) = x^{-1}(1 - \cos x)$$

$$a) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} (De' Hospital) = \lim_{x \rightarrow 0} \frac{\sin x}{1} = 0$$

$$b) \cos x \approx 1$$

$$x \approx 2\pi k, k \in \mathbb{Z}$$

$$1 - \cos x = 2 \sin^2 \frac{x}{2}$$

$$f(x) = \frac{2 \sin^2 \frac{x}{2}}{x} = (t = \frac{x}{2}) = \frac{2 \sin^2 t}{2t} = t \left(\frac{\sin t}{t} \right)^2 = \frac{x}{2} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2$$

$$\lim_{x \rightarrow 0} \frac{\sin t}{t} = 1 \text{ dla } x \rightarrow 0 (t \rightarrow 0) \text{ mamy } f(t) = t 1^2 = t = \frac{x}{2}$$

2.4

$$f(x) = \sqrt{x+2} - \sqrt{x}$$

Problem dla dużych x : $\sqrt{x} \approx \sqrt{x+2}$

$$\sqrt{x+2} - \sqrt{x} = \frac{\sqrt{x+2} - \sqrt{x}}{\frac{\sqrt{x+2} + \sqrt{x}}{\sqrt{x+2} + \sqrt{x}}}$$

$$\frac{x+2-x}{\sqrt{x+2} + \sqrt{x}} = \frac{2}{\sqrt{x+2} + \sqrt{x}}$$

$$a^2 - b^2 = (a-b)(a+b)$$

2.5

$$u = \pm \sqrt{\frac{x + \sqrt{x^2 + y^2}}{2}}$$

$$v = \frac{y}{2u}$$

$$\text{liczba zespolona} = x + iy$$

$$x \neq 0$$

pierwiastek to $u + iv$

$x \geq 0$: jeśli $y \ll x$, to może dojść do "pochłonięcia" y

poza tym dokładność generalnie dobra

$x < 0$: może się zdarzyć, że $|x| \approx \sqrt{x^2 + y^2}$

i nastąpi utrata cyfr znaczących w liczniku, więc dokładność wzoru jest słaba

alternatywny wzór na $k < 0$:

$$x + \sqrt{x^2 + y^2} = \frac{(x + \sqrt{x^2 + y^2})(x - \sqrt{x^2 + y^2})}{(x - \sqrt{x^2 + y^2})} = \frac{x^2 - x^2 - y^2}{x - \sqrt{x^2 + y^2}} = \frac{-y^2}{x - \sqrt{x^2 + y^2}} = \frac{-y^2}{-(|x| + \sqrt{x^2 + y^2})}$$

$$\text{Wtedy } u = \sqrt{\frac{-y^2}{2(x - \sqrt{x^2 + y^2})}} = \sqrt{\frac{y^2}{2(|x| + \sqrt{x^2 + y^2})}}$$

2.6

$$4ac \rightarrow 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 x_2 = \frac{c}{a} \quad q = \frac{1}{2a}(-b - \sqrt{b^2 - 4ac})$$

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \frac{-(b + \sqrt{b^2 - 4ac})}{2a}$$

$$x_1 = \frac{c}{ax_2}$$

$$\begin{aligned}b &= 0ax^2 + c = 0 \\ \Delta &= -4ac\end{aligned}$$

$$\begin{aligned}b &< 0 \\ x_1 &= \frac{|b|+\sqrt{b^2-4ac}}{2a} \\ x_2 &= \frac{c}{2a}\end{aligned}$$

2.8

$$\begin{aligned}\frac{|f(\hat{x})-f(x)|}{|f(x)|} &= \frac{|f(x+x\delta)-f(x)|}{|f(x)|} \approx \frac{|f'(x)x\delta|}{|f(x)|} = cond(x)|\delta| \\ x^\alpha(x^\alpha)' &= \alpha x^{\alpha-1} \frac{|(\alpha x^{\alpha lpha-1})x\delta|}{|x^\alpha|} = \frac{|\alpha x^\alpha \delta|}{|x^\alpha|} = \alpha \delta \\ \sin x(\sin x)' &= \cos x \frac{|(\cos x)x\delta|}{|\sin x|} = \frac{x\delta}{\tan x} \\ e^x(e^x)' &= e^x \\ x^{-1}e^x(x^{-1}e^x)' &= x^{-1}e^x + (-1)x^{-2}e^x = \frac{e^x}{x} - \frac{e^x}{x^2} \\ \arcsin x\end{aligned}$$

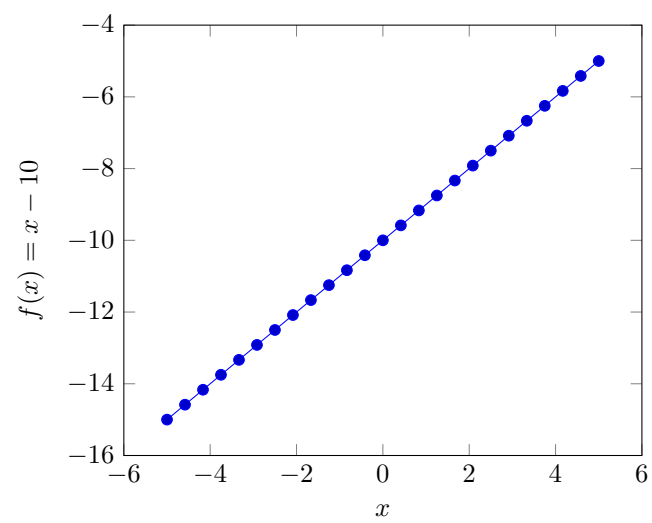
3 Lista 3

3.1

$$\begin{aligned}\text{Bisekcji } [a_0, b_0] \ c_n &= \frac{a_n+b_n}{2} \\ \lim_{n \rightarrow \infty} c_n &= r \\ e_n &= r - c_n\end{aligned}$$

3.1.a)

$$\begin{aligned}f(x) &= x - 10 \\ \text{Czy relacja } |e_0| \geq |e_1| \geq \dots \\ r &= 10 \\ f(10) &= 0 \\ a_0 &= 0 \\ b_0 &= 22 \\ c_0 &= 11 \\ e_0 &= -1 \\ |e_0| &= 1 \\ a_1 &= 0 \\ b_1 &= 11 \\ c_1 &= 5.5 \\ e_1 &= 4.5 \\ |e_0| &< |e_1|\end{aligned}$$



3.1.b)